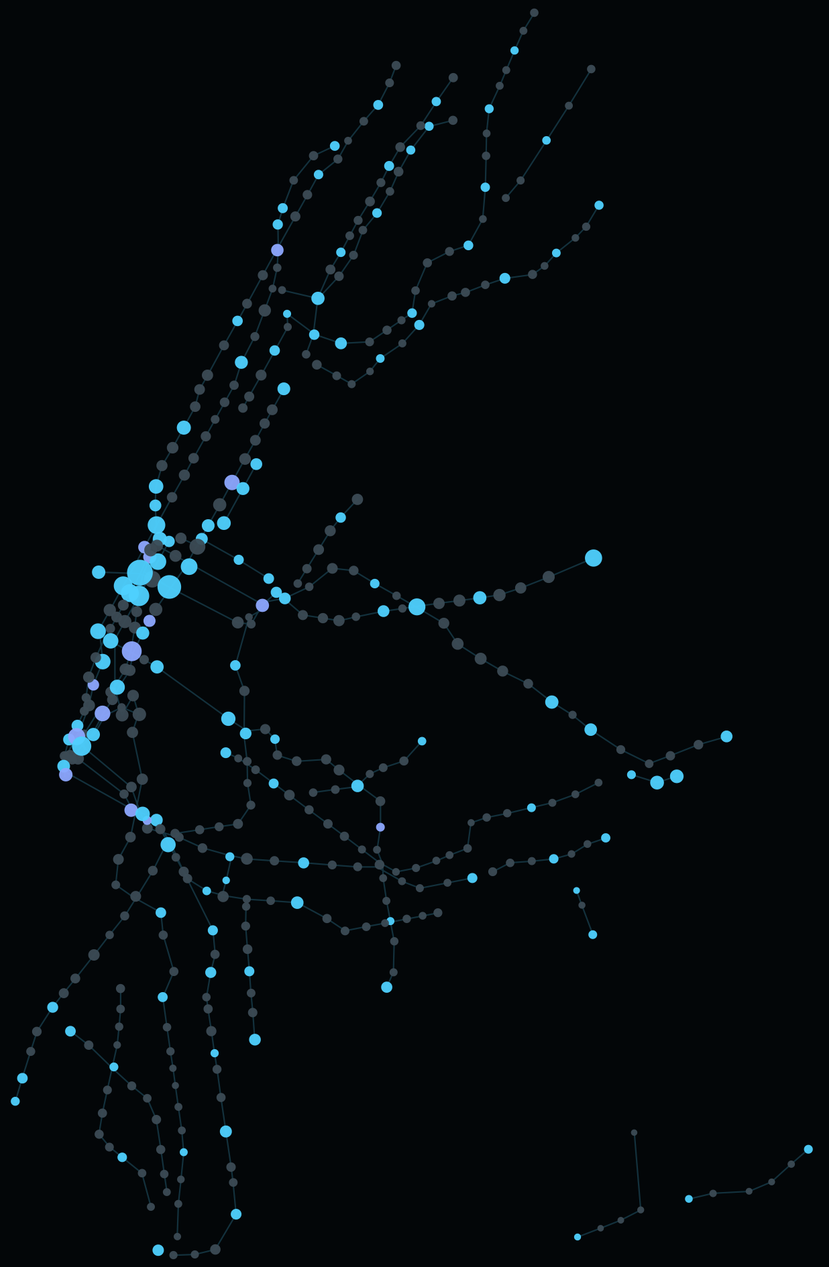
STEPFREE // NYC

Which subway stations should get elevators next?

An optimisation model and planning tool for New York's subway accessibility program.



424 station complexes · 25,820 ridership records · 5 NYC open datasets

Pritish Sharma · Rutgers Data Science Club · NYC in Motion · 7 Nov 2026

01 — THE PROBLEM

Most of the subway still starts with stairs.

296 / 424

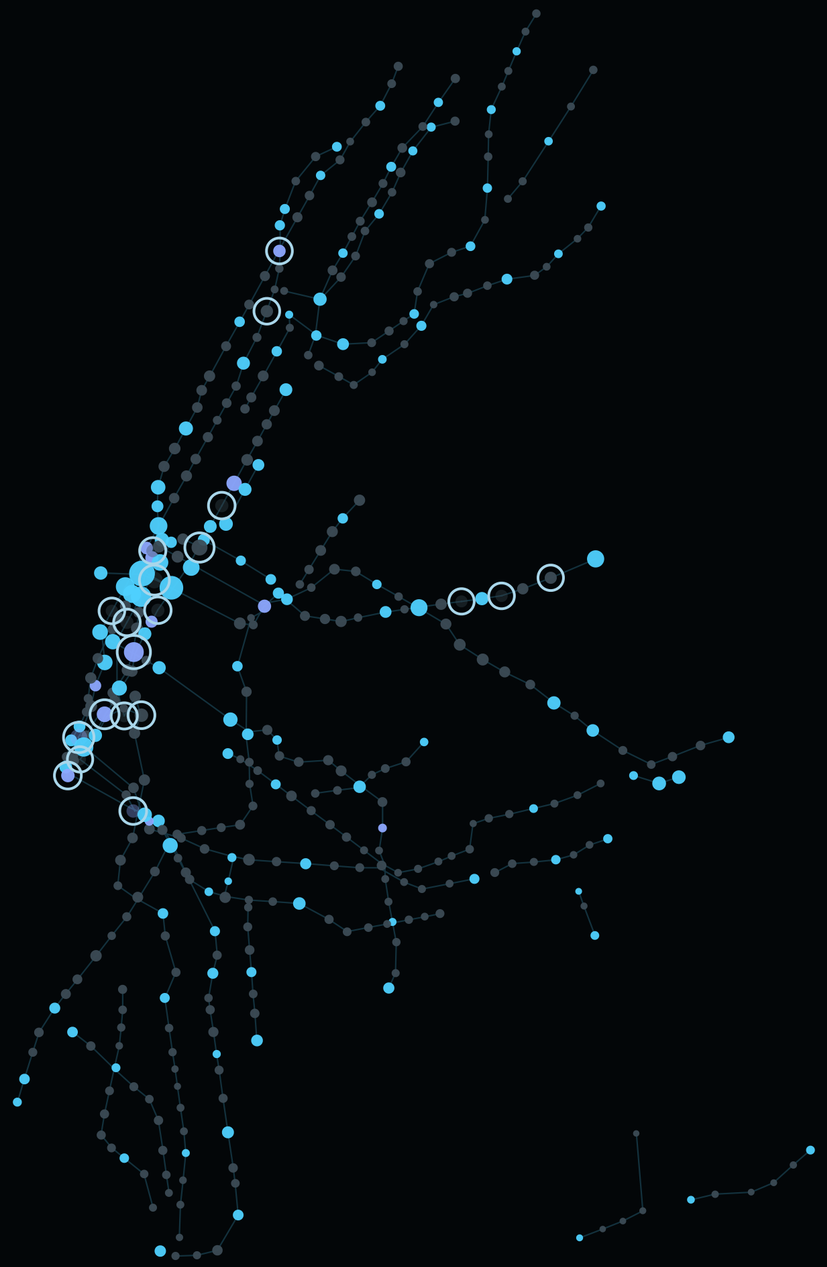
complexes lack step-free access on at least one route

0.97 km

average trip to a step-free station on the same route, for riders on those routes

MTA 2022 settlement → 95% of stations accessible by 2055.
Every capital cycle is a sequencing decision.





02 — THE OBVIOUS RULE

“Upgrade the busiest stations first.”

The 20 busiest inaccessible complexes carry 510,587 weekday entries.

BUSIEST 20 · AVG. DETOUR

0.72 km

MODEL’S 20 · AVG. DETOUR

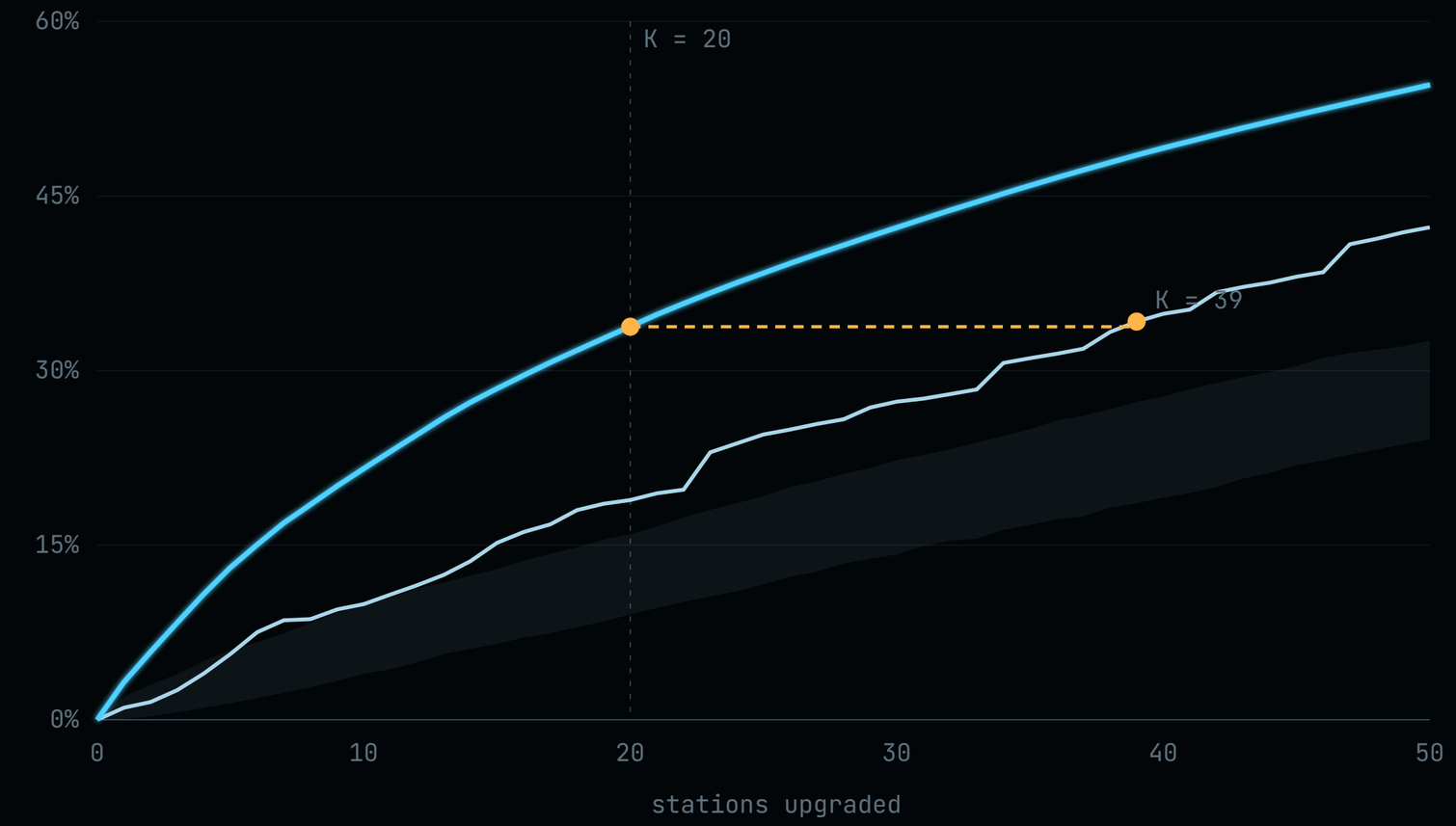
1.26 km

Busy stations often already have a step-free option nearby on the same line. Only 8 of 20 stations appear on both lists.

03 — THE DISCOVERY

Same budget. 79% more of the gap closed.

Share of the Access Gap Index closed · grey band: 500 random orders (5–95%)



BUSIEST-FIRST · 20 UPGRADES

18.9%

STEPFREE OPTIMISER · 20 UPGRADES

33.8%

TO MATCH IT, BUSIEST-FIRST NEEDS

39 upgrades

04 — WHY IT WORKS

Classon Av (G)

Weekday entries	4,080
Rank by ridership	#154

NEAREST STEP-FREE STATION ON THE SAME ROUTE

2.84 km

RANK BY BENEFIT

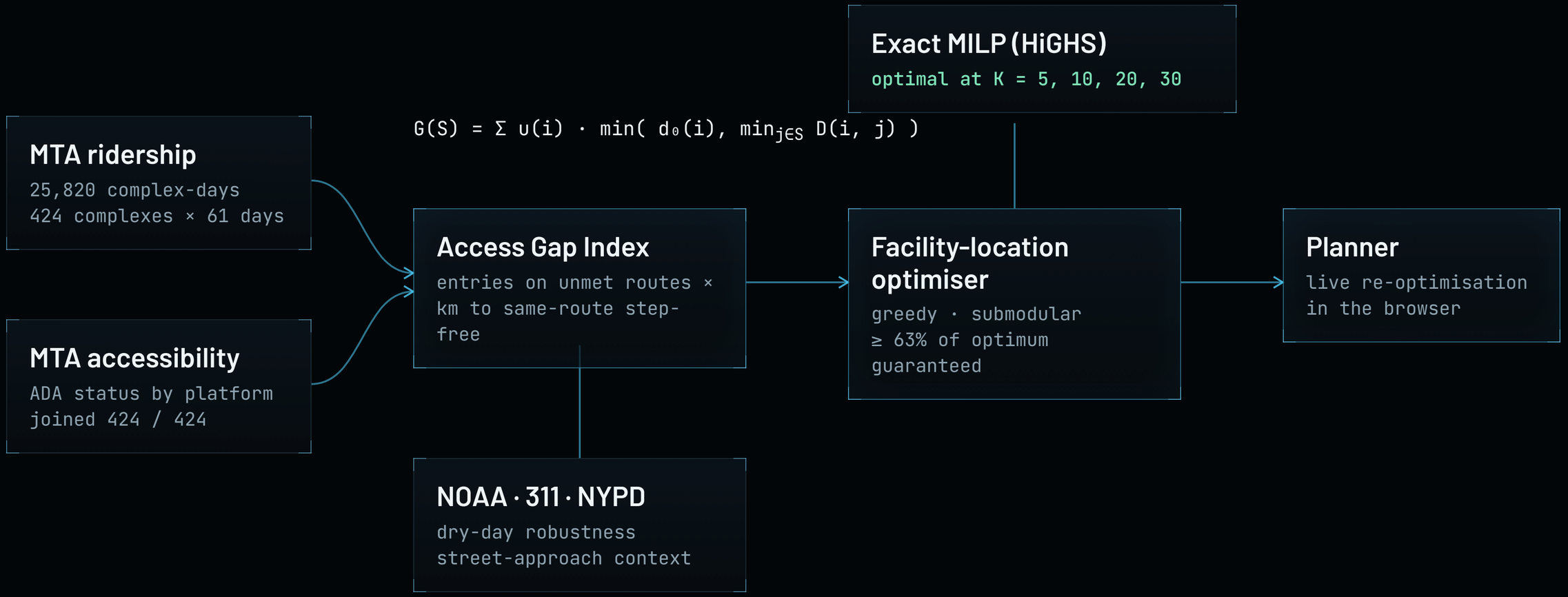
#3

also shortens the trip for 4 neighbours:
Myrtle-Willoughby Avs 2.10 → 1.12 km
Bedford-Nostrand Avs 2.66 → 0.56 km
Clinton-Washington Avs 2.68 → 0.58 km
Fulton St 2.36 → 1.31 km



05 — METHOD

A facility-location model, checked against the exact optimum.



06 — EVIDENCE

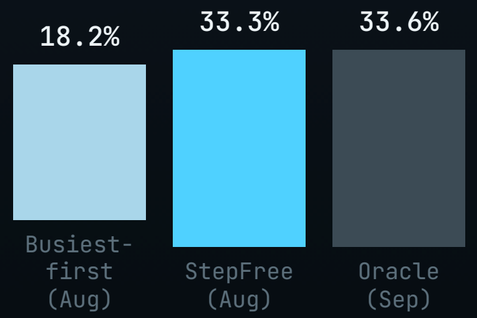
Four tests. The recommendation holds.

OPTIMALITY

4 / 4

budgets where greedy equals the exact MILP optimum (K = 5, 10, 20, 30)

TIME HOLD-OUT



plan built on August weekdays, scored on September

BOOTSTRAP

18 / 20



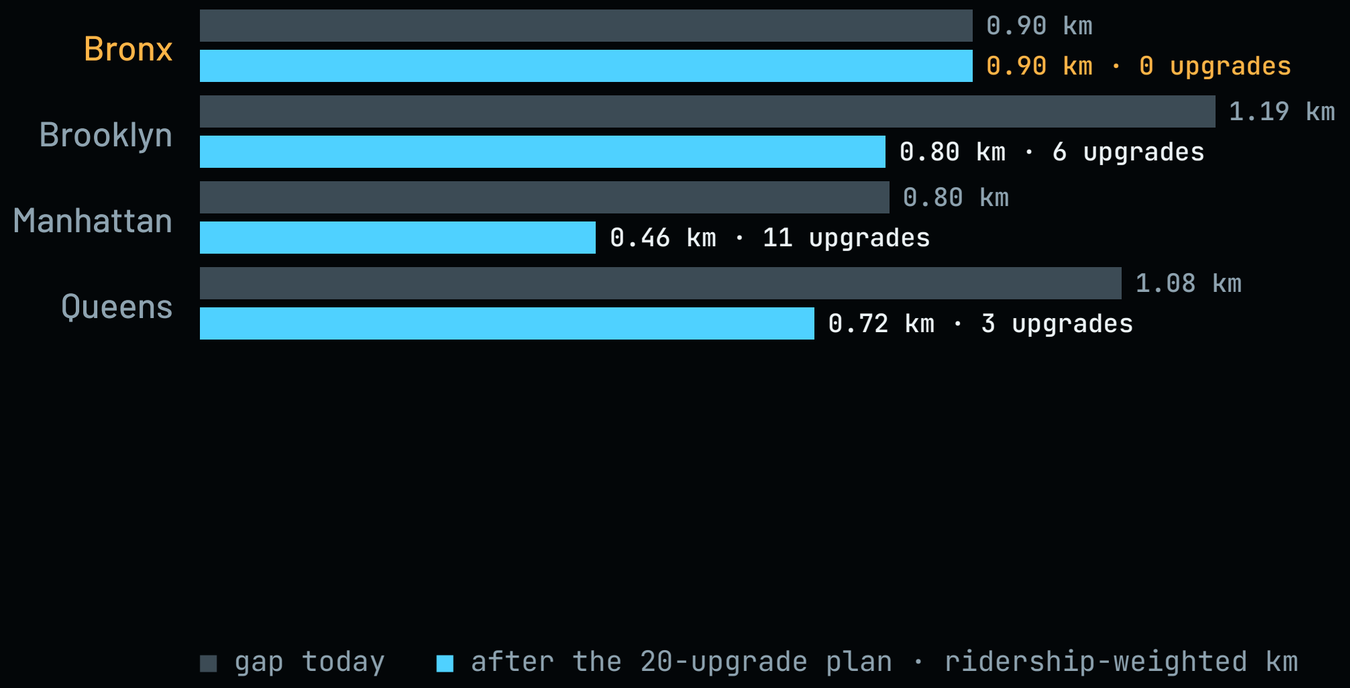
picks kept in ≥ 90% of 200 weekday resamples (dot opacity = frequency)

SENSITIVITY · TOP-20 OVERLAP

Dry weekdays (NOAA)	20 / 20
Cap 3 km / 8 km	20 · 20
Strict partial rule	19 / 20
Weekend demand	16 / 20
Any-route rule (policy)	6 / 20

07 — EQUITY

Pure efficiency gives the Bronx nothing.



BRONX UPGRADES IN THE TOP 20

0

COST OF GUARANTEEING ≥ 1 PER BOROUGH

−0.46 pts

33.8% → 33.3% of the gap closed, with the Bronx receiving 1 upgrade.

08 — THE PLANNER

The model, as a decision tool.

01 Re-solves live as the budget and rules change

02 Explains every rank in plain language

03 Shows the gap before and after on the map

04 Equity floors and engineering exclusions

→ live demo



09 — IMPACT & LIMITS

What it changes, and what the data can’t tell us.

IF THE MTA USED IT

- → A **defensible, auditable** sequence, re-run each capital cycle in seconds
- → Every pick shows its gap, its benefit and **who it helps**
- → Equity floors and engineering exclusions built in
- → Flags where an elevator needs a **DOT curb and sidewalk fix**

WHAT THE DATA CAN’T TELL US

- ! Entries are a **proxy** for who needs step-free access
- ! Straight-line distance; origin side of trips only
- ! No cost data: add it and the same code solves a **budget knapsack**
- ! Accessibility is a recent snapshot; refresh before use

Same budget. Nearly twice the access.

Twenty elevators, placed where they close the most distance: 33.8% of the gap, against 18.9% for the busiest-first rule.

github.com/pritish-sharma/stepfree-nyc · Pritish Sharma

